

Manush Prajwal

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Professional Summary

Software engineer passionate about building scalable backend systems and AI-powered applications using Node.js, NestJS, and AWS. **5X Hackathon Winner** (Google, BNMIT, RNSIT, TOCE) and author of **11 research publications (9 IEEE, 2 journals)**. Founding engineer of **Atlas Bill**, a retail ERP platform serving **26 stores across Karnataka and Tamil Nadu**, generating **INR 104K revenue in 47 days** and processing **₹600K+ monthly transaction value**.

Skills

Backend & Systems: Node.js, NestJS, REST APIs, Microservices, AWS (EC2, RDS, S3, Lambda, CloudWatch), GCP

AI & Agentic Systems: LangChain, LangGraph, PhiData, LangFlow, LLM Orchestration, Agentic Loops

DevOps & Infrastructure: Docker, Terraform, Prometheus, Sentry, CI/CD, Infrastructure as Code (IaC)

Databases: PostgreSQL, Supabase, MongoDB

Languages: Python, TypeScript, JavaScript, SQL, C, C++

Achievements

- **5X Hackathon Winner** — Google, BNMIT, RNSIT, TOCE Certificates | IEEE Xplore
- **3rd Runner-Up** — **Google GenAI Exchange Hackathon 2025** among **278K+ developers** and **4,457 prototypes** submitted.
- **Research Publications** Author of **11 research papers (9 IEEE conference, 2 journals)** across machine learning, distributed systems, and secure computing architectures.
- **Product Impact** Built **Atlas Bill ERP platform**, generating **INR 104K revenue in 47 days** with **26+ stores** onboarded.

Professional Experience

Research Intern — Machine Learning and Photonic Biosensors

Oct 2023 – Jan 2024

IEEE Photonics Society, Bengaluru

- Engineered a diagnostic framework integrating **machine learning with photonic biosensors**, achieving **90.03% accuracy** in diabetes detection.
- Collaborated with **IIT postgraduate researchers**, contributing to published IEEE papers through experimental analysis and model validation.
- Implemented data preprocessing pipelines for biosensor signals, reducing noise by **40%**.
- Presented findings at **INDIACom 2024** and **IEEE ICCCNP 2025**.

Education

The Oxford College of Engineering

Sept 2023 – Present

B.E. in Computer Science and Engineering (**CGPA: 8.3/10**)

Expected Graduation: May 2027

Relevant Coursework: Distributed Systems, Database Management, Operating Systems, Machine Learning

Alvas PU College, Moodubidri

2021 – 2023

Pre-University Course in PCMC (**Percentage: 91.6%**)

Projects

AWS CloudPilot — Autonomous FinOps & Infrastructure Optimizer

[Live | GitHub](#)

- Engineered an autonomous agentic system that monitors cloud infrastructure, detects cost anomalies, and executes safe, deterministic Terraform code, generating infrastructure right-sizing recommendations with estimated cost-saving opportunities of up to **30-40%** in simulated workloads.
- Architected a 4-stage **Agentic Loop**: Monitoring Agent (parses CloudWatch/Prometheus metrics), Reasoning Agent (evaluates guardrails), Planning Agent (generates Terraform scripts), and Execution Agent (sandboxed deployment).
- Implemented cost anomaly detection (identifying idle staging clusters costing **\$120/day**) and automated right-sizing recommendations using **Terraform** and AWS SDK integrations, reducing cloud spend per resource.
- Designed & scaled a secure Terraform execution environment with validation sandbox and **human-in-the-loop validation gates** with automated rollback capabilities, ensuring infrastructure changes never bypass safety checks.

Atlas Bill — Retail ERP Startup Platform

[Live | GitHub](#)

- Led the development and scaled a retail ERP platform powering daily operations for **26 supermarkets across Karnataka and Tamil Nadu**, generating **INR 104K revenue in 47 days of launch**.
- Engineered comprehensive ERP modules for billing, inventory, CRM, analytics, audit tracking, and financial reporting, reducing manual operational effort by **65%**.
- Designed & scaled real-time inventory synchronization with automated billing workflows, processing **INR 600K+ monthly transaction value**.
- Architected backend infrastructure using a **multi-tenant database design** supporting high-scale business transactions, built on **Node.js, PostgreSQL, Docker**.